

Beyond “Prompting”: Redesigning Computer Science Education for the Post-Generative-AI Workforce

Peyman Teymoori 

School of Business, University of South-Eastern Norway

Drammen, Norway (peyman.teymoori@usn.no)

Abstract

The rapid integration of Generative Artificial Intelligence (GenAI) into software engineering is disrupting the traditional entry-level labor market, raising urgent questions for computer science pedagogy. While recent industry reports project significant productivity gains, they also reveal an emerging "hollow middle": the automation of routine debugging and syntax generation tasks that historically served as essential training ground for novice engineers. This article argues that the severance of this "learning by doing" pipeline risks inducing "cognitive debt" in students—producing graduates capable of generating sophisticated code but lacking the deep conceptual understanding required to verify or maintain it. Addressed primarily to the computing education community, this paper adopts an integrative literature review approach to synthesize labor market signals, cognitive science research on the "multiple demand network," and sociological analyses of student practices. From this synthesis, we develop **Cognitive Apprenticeship 2.0 (CA 2.0)**—a theoretical pedagogical framework comprising three pillars: Epistemic Verification, AI Orchestration, and Process Transparency. We then present a proposed validation design, outlining a two-phase pilot implementation within a Bachelor's degree program intended to empirically test the framework's core constructs. Rather than reporting empirical findings, this paper contributes conceptual vocabulary and structured intervention logic to guide future research and curriculum reform. We argue that future-proofing computing students requires a fundamental shift from syntax-centric instruction toward cultivating the systems thinking, evaluative judgment, and tacit knowledge that current AI systems cannot replicate.

Keywords: Computer Science Education and Assessment, Cognitive Apprenticeship, Generative AI, Curriculum Design, Adaptive Expertise, Cognitive Debt.

1. Introduction

1.1 The Pedagogical Crisis of the "Hollow Middle"

For decades, the pedagogical trajectory of a software engineer was mirrored by the structure of the labor market. Novices entered the workforce possessing basic syntactic knowledge, and their early years were defined by codified knowledge tasks: fixing minor bugs, writing unit tests, and implementing well-defined specifications. These tasks served a critical educational function: they constituted the deliberate practice required to build sophisticated mental models of complex systems (Ericsson et al., 1993). However, the advent of Large Language Models (LLMs) and "Agentic AI" systems (McKinsey, 2025) has introduced a discontinuity in this model, severing the mechanism that produces the senior engineers the industry needs.

We are currently witnessing a paradox that strikes at the heart of the Computer Science (CS) curriculum: while AI is dismantling the entry-level roles that graduates rely on for skill consolidation, the demand for high-level architectural judgment remains stable (Brynjolfsson et al., 2025a). Economic projections suggest that AI could expose the equivalent of 300 million full-time jobs globally to automation (Goldman Sachs, 2023). Recent data indicates that 66% of global enterprises plan to reduce entry-level hiring due to AI adoption (Intuition Labs, 2025). For educators, this "hollow middle" presents a specific challenge: *if the "bottom rungs" of the career ladder are automated, universities can no longer rely on industry*

apprenticeships to bridge the gap between academic syntax and professional engineering. The curriculum itself must now bridge this gap.

1.2 Cognitive Atrophy and the "Wrong Kind of AI" in the Classroom

The integration of AI into the curriculum involves risks beyond future employability. As universities rush to adopt AI tools under the banner of "innovation," there is a growing concern regarding the "deskilling" of critical thought (AAUP, 2025). A growing coalition of academics has formally warned against uncritical adoption of AI technologies, arguing that deployment often outpaces understanding of their limitations and societal effects (Guest et al., 2025). Questions of how to regulate AI use in academic settings remain contentious, with scholars noting the tension between innovation imperatives and educational integrity (Colonna, 2022). When students offload the cognitive struggle of coding to an LLM, they accumulate "cognitive debt", a deficit in neural connectivity and deep conceptual understanding that results from bypassing the productive friction of learning (Purser, 2025; Kosmyna et al., 2025).

This aligns with Acemoglu and Restrepo's (2020) warning against the "wrong kind of AI", technology that automates work without creating new tasks. Some scholars argue more starkly that while AI may not fully replace human workers, uncritical integration poses existential risks to the very nature of human cognitive development and professional competence (Xie, 2025). In the educational context, the "wrong kind of AI" usage manifests as students engage in "vibe programming," where code is generated via natural language without an understanding of the underlying logic (Michigan Tech, 2025). This creates a "jagged technological frontier" where students may produce sophisticated outputs while lacking the fundamental competence to verify, debug, or secure them (Dell'Acqua et al., 2023).

1.3 Reframing Learning Objectives: From Coders to Orchestrators

Despite these challenges, the obsolescence of the human software engineer is not a foregone conclusion; rather, the learning objectives for the discipline must shift from "construction" to "orchestration." As AI systems evolve from passive chatbots to autonomous agents capable of planning and executing complex workflows (McKinsey, 2025), the market value of a graduate will shift from syntactic fluency to *evaluative judgment* and *systems thinking*. Therefore, the imperative for higher education is to prepare students not just to use AI, but to audit, govern, and orchestrate it. Scholars have increasingly framed this moment as a 'clarion call' requiring fundamental reconceptualization of both learning objectives and assessment practices (Overono & Ditta, 2025), a shift that requires moving beyond the 'techno-optimism vs. doom' binary toward a pragmatic restructuring of pedagogy (Dubhashi, 2022).

1.4 Research Objectives and Questions

The primary objective of this paper is to bridge the gap between rapidly shifting labor market realities and the relatively static conventions of computer science education. To achieve this, this integrative review is guided by three central research questions:

- **RQ1 (Contextual):** How is the automation of codified knowledge by Generative AI reshaping the structure of entry-level software engineering employment?
- **RQ2 (Cognitive):** What are the cognitive and pedagogical risks associated with "offloading" syntax generation and debugging to AI agents during the novice phase of learning?
- **RQ3 (Pedagogical):** What design principles and curricular frameworks are required to mitigate these risks and create "adaptive expertise" in the post-GenAI era?

In answering these questions, this work makes the following contributions to the field of computing education:

- **Cognitive Debt as a post-GenAI risk construct.** We synthesize labor market signals and cognitive science to link the emerging "hollow middle" in software employment to a specific

educational risk: *cognitive debt*—a widening gap between AI-assisted performance and underlying competence. We define this construct, distinguish it from cognitive load and novice status, and outline behavioral and epistemic indicators that make it observable in classroom assessment.

- **A three-pillar framework mapped to concrete assessment practices.** Extending classical cognitive apprenticeship (Collins et al. 1991) models, we propose **Cognitive Apprenticeship 2.0**, organized around *Epistemic Verification*, *AI Orchestration*, and *Process Transparency*. Each pillar is tied to specific assessment formats—such as adversarial AI-code audits, test-first “prompt-to-product” labs, version-history forensics, and oral verification, offering a structured way to redesign courses for the post-GenAI era.
- **A program-level implementation model integrating AI orchestration into capstone project.** We outline a two-phase pilot (in the “Business Intelligence” and “Bachelor Thesis” courses) in which AI use is not merely permitted but *formally orchestrated and audited* through process logs, methodology chapters, and oral defenses. This provides a portable template for embedding AI governance and verification into high-stakes projects and capstones in other software engineering programs.

1.5 Organization of the Paper

The remainder of this paper is structured to address these guiding research questions sequentially. **Section 2** discusses our methodology. **Section 3** addresses **RQ1** by synthesizing labor market data to define the “Hollow Middle” and the structural displacement of entry-level roles. **Section 4** addresses **RQ2** by exploring the neural and behavioral risks of “cognitive debt” and the critical distinction between augmentation and atrophy. **Section 5** responds to **RQ3** by proposing the Cognitive Apprenticeship 2.0 framework and detailing its pilot implementation in a capstone project. **Section 6** discusses the limitations and our future work. Finally, **Section 7** presents our concluding remarks.

2. Methodology

Given the nascent and rapidly evolving nature of Generative AI, this paper adopts an **Integrative Literature Review** approach (Torraco, 2005). Unlike systematic reviews that aggregate established empirical evidence to answer a narrow query, an integrative review synthesizes diverse streams of literature—experimental, theoretical, and non-empirical—to generate new frameworks for emerging topics. This method is particularly appropriate here, as it allows for the triangulation of established learning sciences theories (e.g., Cognitive Apprenticeship) with real-time labor market signals that have not yet undergone multi-year longitudinal study. Furthermore, the paper presents the design of a pilot intervention to illustrate the practical application of the proposed framework.

2.1 Search Strategy and Selection Process

A comprehensive search was conducted for literature published between November 2022 and October 2025, alongside foundational texts in cognitive science. We utilized a multi-database search strategy targeting three distinct domains: Computer Science Education (ACM Digital Library, IEEE Xplore), Labor Economics (NBER, SSRN, Google Scholar), and Cognitive Science (PubMed, PsycINFO).

To ensure high relevance, the search process followed an iterative screening protocol:

1. **Initial Screening:** Queries such as “Generative AI in Software Engineering,” “Novice Programmers LLM,” and “Labor Market Displacement IT” yielded an initial pool of approximately 200 potential sources.

2. **Relevance Filtering:** Search results were reviewed in rank order. Due to the high volume of generalist AI content, we limited detailed review to the first 30–50 results per query, stopping when relevance diminished (saturation).
3. **Inclusion/Exclusion Criteria:** To manage domain overlap, strict criteria were applied.
 - a. **Inclusion:** Sources were retained only if they specifically addressed *software engineering contexts*, *IT labor market dynamics*, or *cognitive mechanisms relevant to coding*.
 - b. **Exclusion:** Broad discussions of AI in the humanities, general automation studies not specific to knowledge work, non-IT fields, non-knowledge work automation and pre-LLM educational tools were excluded.
 - c. **Result:** This screening process reduced the pool to a final set of 58 sources used in this review.

2.2 Hierarchy of Evidence and Source Types

Given the nascent nature of the topic, traditional peer-reviewed journals have not yet caught up with the velocity of GenAI development. Therefore, this review deliberately includes "gray literature" to capture real-time market signals. We employed a stratified "Best-Evidence" approach:

1. **Peer-Reviewed Empirical Studies (Foundational):** Used for established cognitive principles (e.g., Ivanova et al., 2020) and pre-2023 pedagogical theories.
2. **Academic Working Papers (Emerging):** Sourced from reputable repositories (e.g., NBER, arXiv) to capture recent economic data.
3. **Industry Reports (Contextual):** High-currency reports (e.g., McKinsey, Intuition Labs) are utilized strictly as "market signals" regarding hiring trends, rather than as scientific fact.

2.3 Thematic Analysis and Synthesis

The final selection of 58 sources was analyzed using a deductive-inductive thematic synthesis approach (Torraco, 2005).

- **Coding:** Sources were first coded deductively based on the three research questions (e.g., codes for "displacement," "cognitive load," "assessment").
- **Clustering:** As analysis progressed, inductive codes emerged from the data (e.g., "senior premium," "internal compiler"). These codes were clustered to identify patterns that bridged the disparate domains of economics and cognition.
- **Synthesis:** These clusters formed the three dominant themes that structure this paper: *Structural Displacement*, *Cognitive Divergence*, and *Pedagogical Restructuring*.

Table 1 provides a summary of the distribution of sources used in this synthesis, highlighting the balance between established theory and emerging market data.

Table 1: Distribution of Literature Sources by Domain and Type

DOMAIN	PRIMARY FOCUS	SOURCE INCLUDED	TYPES	COUNT (APPR.)	ROLE IN REVIEW
Labor economics & industry	Market displacement, hiring trends, productivity metrics.	Industry Reports (e.g., McKinsey), NBER Working Papers, Economic Analysis.		14	Establishing the "Hollow Middle" phenomenon (RQ1).
Cognitive science	Neural mechanisms of coding, memory, and cognitive load.	Peer-Reviewed Journals (e.g., <i>Nature</i> , <i>eLife</i>), Psychology Reviews.		12	Defining the "Multiple Demand" network and cognitive risks (RQ2).
CS education (CSed)	Pedagogy, assessment, student practices with AI.	Conference Proceedings (ACM, IEEE), Journal Articles, Preprints.		24	Analyzing student behavior and proposing CA 2.0 (RQ3).

Policy & ethics	Institutional governance, academic integrity.	Association Reports (AAUP), Position Papers, Essays.	8	Contextualizing the ethical implementation of the framework.
Total sources				58

Note: The timeframe of selected literature is heavily weighted toward the post-GenAI era (2022–2025), with foundational cognitive science texts dating prior to 2022.

This triangulation ensures that the proposed pedagogical interventions are not merely reactive but grounded in both economic reality and cognitive science.

3. The Structural Disintegration of the Entry-Level Pipeline

The integration of Generative AI (GenAI) into the software development lifecycle has not resulted in a uniform uplift in productivity across all experience levels. Instead, empirical data from 2024 and 2025 indicates a fracturing of the traditional labor pyramid. We identify this phenomenon as the **“Hollow Middle”**—a market condition where demand for high-level architectural expertise remains robust, while the entry-level roles that historically served as the training ground for that expertise are systematically automated. Given the recency of GenAI, this analysis relies on real-time employer surveys and working papers; we therefore treat the ‘hollow middle’ as a *strong emerging signal* rather than a settled historical fact.

3.1 Empirical Evidence of the "Hollow Middle"

Theoretical debates regarding AI’s impact on employment are often speculative; predictions dating back a decade anticipated significant disruptions to labor markets from AI and robotics (Smith & Anderson, 2014). However, recent industry data provides early signals of a structural shift. While long-term peer-reviewed economic data is still crystallizing, real-time hiring metrics reveal a crisis specifically targeting early-career professionals. The World Economic Forum projects that 39% of core workplace skills will change by 2030, with *analytical thinking* and *AI literacy* among the fastest-growing skill requirements (World Economic Forum, 2025).

Multiple independent data sources provide converging evidence of a structural shift in entry-level employment. The pace of this transformation is historically unprecedented; survey data indicates that ChatGPT achieved faster adoption than any prior technology, with approximately 40% of the U.S. population aged 18-64 used Gen AI by late 2024, with 23-24% using it for work weekly (Bick et al., 2024). Analysis of ADP payroll records covering millions of U.S. workers reveals a 13% relative decline in employment for early-career workers (ages 22–25) in AI-exposed occupations since late 2022, even after controlling for firm-level shocks (Brynjolfsson et al., 2025a). Federal Reserve Bank of New York data indicates the unemployment rate for recent college graduates reached 5.8% in Q1 2025, the highest since 2021, with the gap between graduate and overall unemployment at a historic low (HubbardOBrienEconomics, 2025). Job posting data from Revelio Labs shows entry-level postings declined 35% from January 2023, with AI-exposed entry-level roles experiencing steeper declines than non-exposed roles (Simon, 2025). Research estimates that approximately 80% of the U.S. workforce could have at least 10% of their tasks affected by LLMs, with software developers among the most exposed professions (Eloundou et al., 2024). Similarly, SignalFire's analysis of LinkedIn data documents a 50% decline in new graduate hiring at major tech firms since 2019 (SignalFire, 2025).

3.2 The Agentic Shift: From Chatbots to Autonomous Systems

The displacement of junior roles is accelerating due to the rapid evolution of AI capabilities from passive "chatbots" to active "agents." McKinsey’s *State of AI in 2025* report highlights a critical transition toward **Agentic AI**, systems capable of planning, executing, and refining complex workflows with minimal human intervention. Their data indicates that by 2025, nearly one-third of surveyed

organizations had begun scaling AI systems, particularly in IT and software engineering functions (McKinsey & Company, 2025).

Unlike previous iterations of GenAI that required a human to prompt every step (e.g., "write a function to sort this list"), agentic systems can be assigned high-level objectives (e.g., "refactor this codebase to improve latency"). These agents autonomously break down the goal, generate code, run tests, and iterate over errors. This capability directly overlaps with the primary value proposition of a junior developer. Consequently, organizations are moving toward a model where **AI serves as the "first author"** of code, while human seniors act merely as reviewers and verifiers (Grothe-Hammer et al., 2025).

3.3 The Productivity Paradox: The "Equalizer" Effect vs. The Senior Premium

It is important to acknowledge that within the context of *task completion*, AI acts as a powerful leveler. Contrary to the narrative of pure displacement, empirical studies on tools like GitHub Copilot suggest that AI assistants benefit novice developers disproportionately. Peng et al. (2023) found that less experienced developers saw greater relative productivity gains than their senior counterparts, effectively narrowing the performance gap for routine coding tasks. By bridging the distance between intent and syntax, these tools allow juniors to contribute to production-level codebases much earlier in their careers (George, 2024).

However, this "equalizer" effect generates a paradoxical economic risk: the **"Senior Premium."** While AI allows a junior developer to perform at a higher level, it simultaneously allows a senior developer to act as a "10x engineer," automating the tasks they previously delegated to juniors. If a senior engineer equipped with AI can generate, test, and deploy code at a velocity that negates the need for support staff, the economic rationale for hiring juniors—typically based on lower labor costs for lower-value work—evaporates. The "jagged technological frontier" described by Dell'Acqua et al. (2023) suggests that while novices are more productive *with* the tool, they remain unable to operate effectively *outside* the tool's competence envelope, creating a workforce that is highly efficient but dangerously fragile.

4. The Cognitive Cost of Automation: Atrophy and the Illusion of Competence

The structural displacement of entry-level roles described in the previous section precipitates a second, perhaps more insidious crisis: a fundamental alteration in the cognitive development of novice engineers. As students and junior professionals increasingly rely on AI to generate code, they bypass the "productive struggle" historically required to build mental models of computational logic. This phenomenon, characterized by recent scholarship as "Cognitive Debt" is distinct from "cognitive load" (the immediate mental effort required for a task). Cognitive Debt describes a cumulative deficit in foundational understanding. It threatens to produce a generation of technologists capable of generating sophisticated outputs while possessing brittle, superficial understanding. Operationally, a student with high cognitive debt manifests as one who can rapidly produce working applications via prompting but fails to diagnose logic errors or explain the execution flow without external assistance (Purser, 2025).

4.1 Defining and Operationalizing "Cognitive Debt"

Given the centrality of this construct to our framework, we offer a formal definition to guide both pedagogical application and future empirical validation.

Definition: *Cognitive Debt* is the cumulative deficit in foundational conceptual understanding that accrues when learners repeatedly bypass effortful cognitive processing by offloading problem-solving to external agents (in this case, GenAI). Unlike *cognitive load*, which describes momentary mental effort during a task, cognitive debt is longitudinal and compounding—it represents knowledge and neural development that was never acquired, creating a widening gap between apparent performance and actual competence.

We conceptualize cognitive debt as manifesting across three interrelated dimensions, each with distinct observable indicators:

<i>Dimension</i>	<i>Definition</i>	<i>Observable Indicators</i>
<i>Neural</i>	Reduced development of the cognitive infrastructure (particularly the Multiple Demand network) required for systematic reasoning, resulting from insufficient effortful practice.	Decreased activation in fronto-parietal regions during unassisted problem-solving tasks; reduced EEG connectivity in frequency bands associated with memory encoding (cf. Kosmyna et al., 2025).
<i>Behavioral</i>	Inability to perform foundational programming tasks—such as debugging, tracing execution flow, or identifying logic errors—without AI assistance.	Failure to complete debugging tasks under "AI-off" conditions; inability to explain code logic during oral examination; reliance on trial-and-error prompting rather than systematic diagnosis.
<i>Epistemic</i>	Gaps in declarative and procedural knowledge of core computational concepts, resulting from skipped instruction or bypassed practice.	Poor performance on closed-book conceptual assessments; inability to recognize why AI-generated code is incorrect; absence of accurate mental models when asked to predict program behavior.

These three dimensions are theoretically distinct but empirically correlated: repeated behavioral offloading (Behavioral) prevents the acquisition of foundational knowledge (Epistemic) and the neural consolidation required for expertise (Neural). For the purposes of this framework, we focus primarily on the **Behavioral** and **Epistemic** dimensions, as these are directly accessible through pedagogical assessment. The Neural dimension, while theoretically grounding, requires neuroimaging methodologies beyond typical educational research.

Crucially, cognitive debt is not equivalent to *novice status*. All learners begin with knowledge gaps; what distinguishes cognitive debt is that these gaps persist or widen *despite apparent task completion*, because AI-mediated success masks the absence of underlying learning. A student exhibiting high cognitive debt may produce sophisticated artifacts while remaining unable to reproduce, explain, or adapt the underlying logic independently.

4.2 Neural Divergence: The "Multiple Demand" Network vs. Language Processing

To understand the risk of deskilling, one must examine the neural mechanisms of programming. Established fMRI research has demonstrated that code comprehension activates the **Multiple Demand (MD) network**—a fronto-parietal system essential for complex problem-solving (Ivanova et al., 2020). Writing code manually exercises this network, forging the neural pathways necessary for systemic reasoning.

However, recent preliminary investigations suggest that reliance on AI agents may alter this cognitive engagement. A working paper from the MIT Media Lab (Kosmyna et al., 2025) utilized electroencephalography (EEG) to measure brain activity during AI-assisted tasks. Their initial findings suggest a significant reduction in neural connectivity—specifically in frequency bands associated with memory encoding—when participants relied heavily on AI assistants compared to working unaided. While this represents a small-scale, early investigation, it raises important hypotheses about how AI assistance modulates neural engagement. Critically, these physiological findings converge with established learning sciences theory: specifically, Bjork’s concept of "desirable difficulties" (1994). By removing the friction of syntax generation, AI tools may inadvertently eliminate the cognitive struggle required for deep memory consolidation, potentially impeding the formation of deep conceptual knowledge.

4.3 "Vibe Programming" and the Shift to Editorial Roles

This neural disengagement manifests pedagogically in the rise of "**Vibe Programming**"—an informal paradigm where developers describe desired outcomes in natural language ("vibes") and rely on stochastic models to handle implementation details (Michigan Tech, 2025). While this lowers the barrier to entry, it fundamentally changes the learner's role from "author" to "editor."

A qualitative study of university students in 2025 reveals that this shift is already entrenched. Grothe-Hammer et al. (2025) identified distinct typologies of student AI usage, most notably "**AI as First Author**," where students generate text or code via AI and merely tweak it for submission. In this workflow, the student acts as a "second author," refining output they did not conceive. The danger lies in the **Verification Gap**: students often lack authoritative competence (knowledge of syntax, security, and efficiency) to effectively judge the quality of the AI's output. As Purser (2025) argues, this creates a dynamic where "students graduate fluent in prompting, but illiterate in judgment," effectively outsourcing their cognition to corporate infrastructure.

4.4 The Illusion of Competence and "Brownfield" Development

Cognitive offloading is not inherently detrimental; indeed, the history of computer science is a history of successful abstraction. Just as compilers freed programmers from the cognitive load of memory management, LLMs free them from the cognitive load of syntax recall. Proponents argue that this offloading allows students to focus on higher-order problem solving and "computational thinking" rather than the minutiae of semicolons (George, 2024). Under this model, AI serves as a scaffold that supports learners in tackling problems that would otherwise be beyond their Zone of Proximal Development.

However, the distinction lies in *what* is being offloaded. Unlike a compiler, which is *deterministic*, GenAI is *probabilistic*. When a student offloads the logic generation to an LLM, they are not merely abstracting complexity; they are bypassing the neural activity required to verify truth. The immediate gratification of working code creates a dangerous "illusion of competence." Students using AI tools can produce applications far exceeding their actual skill level, masking their inability to perform foundational tasks unaided. This becomes critical in what software engineering terms "**brownfield development**", maintenance of existing, legacy codebases.

While AI accelerators allow novices to write *new* (greenfield) code rapidly, maintenance requires a deep, mental simulation of how code executes, a skill developed specifically through the manual debugging processes that AI now automates (Vaithilingam et al., 2022). By skipping the frustration of syntax errors and logic bugs, students accumulate cognitive debt that comes due when they encounter edge cases the AI cannot solve. If the academic sector normalizes this dependency under the guise of "innovation," it risks validating a low-quality degree, credentialing students for outputs they did not create and cannot maintain (Purser, 2025).

5. Redefining Pedagogy: Cognitive Apprenticeship 2.0

To mitigate the risks of the "hollow middle" and "cognitive debt," higher education must undergo a fundamental restructuring. The traditional "bootcamp" mentality, focused on rapid syntax acquisition, is no longer sufficient. Instead, we propose a **Cognitive Apprenticeship 2.0 (CA 2.0)** model.

Rooted in the original theory by Collins et al. (1991), which emphasized modeling, coaching, and scaffolding, CA 2.0 adapts these principles for a fundamentally different epistemological environment. In the classic model, the "expert" is a human mentor whose thought processes are visible and trustworthy. In the post-GenAI era, the "expert" generating the artifact is often a probabilistic AI—a "black box" that mimics competence without comprehension. Therefore, CA 2.0 necessitates a shift from imitation to interrogation. While Collins et al. focused on helping apprentices internalize the

expert’s tacit knowledge (Polanyi, 2009), CA 2.0 focuses on helping apprentices verify and orchestrate the output of synthetic intelligence.

5.1 The Three Pillars of CA 2.0

The framework is built upon three pedagogical pillars designed to foster *Adaptive Expertise*—the ability to apply skills flexibly in novel situations (Hatano & Inagaki, 1986). This construct distinguishes between routine expertise, which enables efficient execution of familiar procedures, and adaptive expertise, which enables innovation and flexible response to novel problems (Schwartz, Bransford, & Sears, 2005). These pillars map specific cognitive shifts to actionable teaching strategies, as detailed in Table 2.

Table 2: The Cognitive Apprenticeship 2.0 Framework

PILLAR	SHIFT IN FOCUS	LEARNING OBJECTIVE	PEDAGOGICAL INTERVENTION (ACTIVITY)
1. Epistemic Verification	From <i>Generating</i> to <i>Auditing</i>	Cultivating the "internal compiler" required to validate probabilistic AI outputs and detect hallucinations.	The Adversarial Audit: Students receive a pre-generated AI codebase containing subtle logic errors and security vulnerabilities. The task is not to write code, but to document, explain, and correct the AI's failures without running the executable.
2. AI Orchestration	From <i>Syntax</i> to <i>Systems</i>	Developing the architectural judgment to decompose complex problems into discrete tasks for AI agents.	"Prompt-to-Product" Labs: Students act as "Technical Leads." They must write a comprehensive test suite (TDD) first, then orchestrate AI agents to generate code that passes those tests, effectively grading the AI's performance against their specifications.
3. Process Transparency	From <i>Product</i> to <i>Process</i>	Ensuring the mental model belongs to the student, not the LLM, by assessing the <i>evolution</i> of the solution.	Version History Forensics: Submission requires the full Git history and a "Process Log." Assessment focuses on the decision points: why a specific AI suggestion was rejected, how a prompt was refined, and how the student debugged the integration.

5.2 Positioning CA 2.0 within Existing CS Education Frameworks

Cognitive Apprenticeship 2.0 builds on, but also departs from, several established strands in computing education: structured programming pedagogies such as Predict, Run, Investigate, Modify and Make (PRIMM) (Sentence et al., 2019), collaborative practices such as pair programming (Preston, 2005) and Test-Driven Development (TDD), constructive alignment models (Cain, 2013), and recent “AI across the curriculum” (Southworth et al., 2023) initiatives. Early implementations of AI-integrated CS courses, such as Harvard’s CS50, have demonstrated both the potential and challenges of embedding AI assistants directly into introductory programming instruction (Liu et al., 2024).

5.2.1 Relation to Existing Programming Pedagogies

Approaches like PRIMM help students move from reading and predicting code to modifying and writing it. Pair programming emphasizes collaboration and shared problem solving. CA 2.0 reinterprets these practices under post-GenAI conditions: students are no longer only authors of code but also *auditors* of code produced by opaque systems. Epistemic Verification turns “predict–run–investigate” (as in PRIMM) into a structured *forensic audit* of AI-generated artefacts; AI Orchestration extends pair-programming ideas to a triad of student, AI agent, and human supervisor; and TDD is repositioned as a control surface over generative models, where tests define the specification against which AI outputs are interrogated.

5.2.2 Relation to Constructive Alignment and AI-era Assessment

Work on constructive alignment and on assessment in the age of AI has argued for closer coupling between learning outcomes, tasks, and assessment formats, and has highlighted the value of oral, process-revealing assessments when products may be AI-generated (Farazouli et al., 2024). CA 2.0 makes this alignment discipline-specific: the three pillars map directly to concrete mechanisms—adversarial audits and oral defenses (Epistemic Verification), test-first AI labs (AI Orchestration), and process logs and version-history forensics (Process Transparency). Rather than treating these as ad hoc safeguards, the framework treats them as primary evidence for or against the presence of “cognitive debt.”

5.2.3 Relation to “AI Across the Curriculum”

AI-across-the-curriculum models aim to embed broad AI literacy, ethics, and civic understanding across programmes (Southworth et al., 2023). CA 2.0 is narrower but deeper. It targets a specific failure mode in software engineering: students who can ship impressive AI-assisted systems but cannot independently understand, verify, or maintain them. In that sense, AI-across-the-curriculum provides a horizontal layer of AI literacy, while CA 2.0 offers a vertical, high-stakes layer that redesigns core projects and capstones so that AI becomes a visible, auditable part of the learning process.

5.2.4 Distinctive Contribution

Taken together, CA 2.0 contributes (a) the construct of *cognitive debt* as a way to reason about hidden gaps between AI-assisted performance and underlying competence; (b) a three-pillar framework that translates this risk into specific assessment practices; and (c) a model for integrating AI orchestration, documentation, and verification into the governance of capstone and thesis work. It is best understood as a post-GenAI evolution of cognitive apprenticeship in which the central student role is to *interrogate and orchestrate synthetic expertise*, not merely to consume it.

5.3 The Teacher-AI Task Spectrum: From Sage to Architect

The integration of AI necessitates a re-evaluation of the division of labor in the classroom. According to Koh et al. (2023), educational tasks exist on a spectrum ranging from procedural to high-cognitive. As AI encroaches on lower-order tasks (generating boilerplate), the instructor must shift from the “Sage on the Stage” to a **Systems Architect** of learning experiences.

If AI serves as a “second author” for students (Grothe-Hammer et al., 2025), the instructor’s primary role becomes teaching the *integration* of that work. This aligns with Hernholm (2025), who argues that “uniquely human skills” such as ethical judgment and problem-scoping must move from “soft skills” to core technical differentiators.

5.4 Beyond “Prompt Engineering”: Critical AI Literacy

A dangerous trend in current institutional responses is the reduction of AI education to “prompt engineering”, teaching students merely how to operate commercial tools efficiently. As Purser (2025) critiques, this produces graduates who are “fluent in prompting but illiterate in judgment.” CA 2.0 rejects this functionalist approach. Instead, it mandates **Critical AI Literacy**, where students interrogate the material realities of the tools. Research indicates that learners often hold naive conceptions of AI capabilities that must be explicitly addressed in curriculum (Heeg et al., 2024). This includes understanding issues of trustworthiness and responsible AI modeling (Sarker, 2024), the “Anatomy of Slop” (analyzing failure modes and bias) and recognizing that LLMs are “stochastic parrots” capable of confident hallucinations (Birhane, 2022), not oracles of truth.

The “Human-in-the-Loop” Debugger: Critical literacy also involves recognizing the AI’s limitations in specific technical contexts. Here, the pedagogical value lies in the student’s ability to recognize *when* the AI has failed—such as detecting hallucinated columns in a CSV file or correcting

invalid SQL joins—and intervening. This reinforces the necessity of the "Internal Compiler"; without a fundamental understanding of the underlying logic, students would be unable to prompt the AI to correct its own errors.

The overall structure of CA 2.0 and its comparison with Cognitive Atrophy is illustrated in Figure 1.

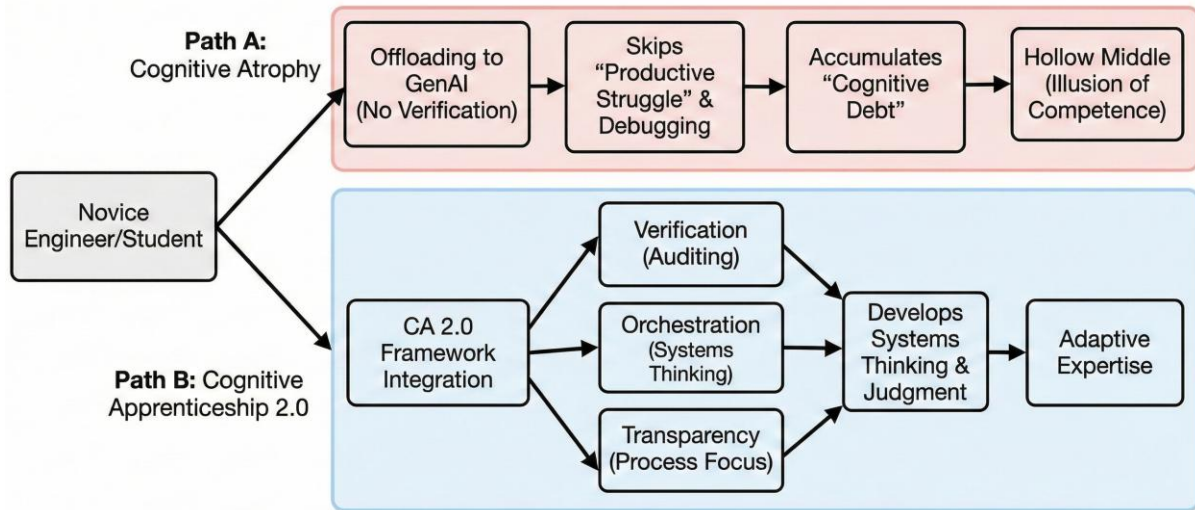


Figure 1 Caption: *The Bifurcation of Competence. Path A (Cognitive Atrophy) illustrates the cycle of offloading without verification, leading to the "Hollow Middle." Path B (Cognitive Apprenticeship 2.0) illustrates the intervention strategy, where the three pillars (Verification, Orchestration, Transparency) serve as the bridge to "Adaptive Expertise."*

5.5 Iterative Implementation: From Coursework to Capstone

To validate the Cognitive Apprenticeship 2.0 framework, we have adopted an iterative implementation strategy spanning two distinct pedagogical contexts at the University of South-Eastern Norway. This approach allows us to observe student "AI Orchestration" behaviors in a technical course (BID3000) before subjecting them to rigorous assessment in the high-stakes Bachelor Thesis (BOP3000).

5.5.1 Phase 1: Exploratory Application

The initial principles of CA 2.0 were introduced in the *Business Intelligence & Data Warehousing* (BID3000) course in Fall 2025. In this technical course, students were tasked with building complex ETL (Extract, Transform, Load) pipelines. Rather than banning AI, students were encouraged to utilize LLMs as "Senior Architects." Analysis of student submissions reveals that high-performing groups spontaneously adopted the **Epistemic Verification** and **Orchestration** pillars:

- **Optimization over Generation:** One cohort utilized AI not to generate boilerplate code, but to refactor inefficient row-by-row SQL insertions into bulk-loading methods, explicitly documenting the performance gain (e.g., "Optimization from 10 minutes to 10 seconds").
- **Concept Generation:** Students used LLMs to generate domain-specific business questions (KPIs) to guide their schema design, effectively using the AI as a "client proxy" rather than a code generator.
- **Provenance Disclosure:** Several groups voluntarily included an "AI Assistance Disclosure" chapter in their technical reports. They detailed specific prompts (e.g., "refine DAX queries") and explicitly stated that "all outputs were reviewed, rewritten, and validated by the group."

These behaviors confirm that students are capable of the "System Thinking" required by CA 2.0 when the assessment criteria value *architecture* over *syntax*.

5.5.2 Phase 2: Rigorous Validation

Building on the successful behaviors observed in BID3000, the Spring 2026 Bachelor Thesis course (15 ECTS) will formalize these practices into mandatory assessment criteria. While BID3000 allowed for voluntary disclosure, the Capstone implementation enforces the **Process Transparency** pillar via three mandatory mechanisms designed to measure "Cognitive Debt":

1. **The Forensic Process Log ("Dagbok"):** Students must maintain a living document recording not just *what* code was written, but *how* GenAI was prompted. This emphasis on structured reflection as a pedagogical tool is supported by emerging research on integrating reflective practices with AI use in higher education (Sellnow, 2025). Crucially, students are required to document '**Failure Cases**'—instances where the AI hallucinated, provided inefficient logic, or failed to connect to database drivers. The student must explicitly write out the logic they used to intervene and correct the AI. This shifts the grading focus from the final artifact to the forensic history of the problem-solving process.
2. **Reflective Methodology Chapter:** The final thesis report (30–50 pages) must include a distinct chapter on "AI Development Methodology." This requirement draws on the tradition of reflective inquiry as foundational to deep learning (Dewey, 1910). Unlike standard method chapters, this must critically analyze the *utility and limitations* of the specific agents used, effectively auditing the AI as a team member.
3. **The Verification Defense:** During the final oral examination, the assessment criteria explicitly weight the student's ability to explain and defend code segments generated by AI. This ensures that while the student may not be the "first author" of the syntax, they possess the "**epistemic sovereignty**" required to verify it.

The BOP3000 pilot is designed to test three core hypotheses:

- **H1 (Process Transparency → Verification Performance):** Students who maintain higher-quality process logs (measured by documented AI failure cases and intervention rationale) will demonstrate stronger performance in oral defense verification questions.
- **H2 (Orchestration Behavior → Independence):** Students exhibiting "AI as Architect" behaviors (using AI for optimization, review, and concept generation rather than wholesale code generation) will show greater ability to modify or debug their code without AI assistance.
- **H3 (Cognitive Debt Detection):** The three-mechanism assessment structure (process log + methodology chapter + oral defense) will identify students whose final artifacts exceed their demonstrable independent competence.

Measurable Outcomes:

INDICATOR	SUCCESS THRESHOLD	FAILURE THRESHOLD
Oral defense pass rate on AI-generated code explanation	≥ 80%	< 60%
Correlation between log quality and defense performance	≥ 0.4	< 0.2
Students unable to explain core code segments	< 15%	> 30%

5.5.3 Data Collection Strategy

To empirically validate this two-phase implementation, future research will correlate the "AI Disclosure" depth from Phase 1 with the "Verification Performance" in Phase 2. We aim to determine if the habit of documenting AI usage (Process Transparency) serves as a predictor for a student's ability to defend their code (Adaptive Expertise).

Although the concrete implementation sketched above is anchored in the local structures of USN (e.g., BID3000/BOP3000, 15 ECTS courses, dagbok requirements, and a bachelor thesis with oral defense),

the Cognitive Apprenticeship 2.0 framework is not specific to a single institution. It is better understood as a set of portable design principles that can be instantiated in many ways depending on program structure, assessment traditions, and regulatory environments.

At a high level, we can distinguish between **context-dependent elements** and **portable principles**.

Context-dependent elements include:

- Credit systems and workload units (e.g., 15 ECTS vs. 3–4 credit modules).
- Formal thesis structures (e.g., whether a capstone is required, length of project).
- Assessment traditions (e.g., prevalence of oral exams or vivas, institutional rules for group vs. individual grading).
- Local tools and platforms (e.g., specific LMS, version control policies, or institutional AI guidelines).

Portable principles of CA 2.0 include:

- **Epistemic verification:** assessment practices that require students to reconstruct, explain, or debug work—whether produced by themselves, peers, or AI tools—under conditions where retrieval cues and external scaffolds are limited.
- **AI orchestration:** explicitly designing tasks where AI tools are treated as fallible collaborators whose suggestions must be evaluated, adapted, or rejected, rather than as infallible oracles.
- **Process transparency:** systematically capturing the trajectory of work (e.g., process logs, version histories, AI-use statements, reflection notes) and making that trajectory visible in assessment.

These principles can be realized across a wide range of course types and levels, even in institutions without extensive capstone projects or oral defenses.

7. Conclusions and Future Research

The integration of Generative AI into the information technology sector represents a rupture in the continuity of professional development. This integrative review set out to analyze this rupture through the lens of labor market shifts, cognitive science, and curriculum design. Returning to our guiding research questions, the early evidence supports three critical conclusions.

7.1 Addressing RQ1: The Structural Reality of the "Hollow Middle"

Regarding the impact on employment (RQ1), our analysis suggests that we are not merely witnessing a cyclical fluctuation in hiring, but potential signs of a structural dissolution of the "junior" role. The empirical signal specifically the 13% decline in early-career employment in AI-exposed fields alongside a sharp reported drop in entry-level job postings (Brynjolfsson et al., 2025b; Intuition Labs, 2025) provide strong market indicators that the "hollow middle" is a developing reality. The automation of routine coding tasks appears to be severing the traditional apprenticeship model, meaning higher education can no longer rely on industry entry-level roles to consolidate foundational skills.

7.2 Addressing RQ2: The Danger of Cognitive Atrophy

Regarding the risks of AI reliance (RQ2), the findings suggest a bifurcation of paths: "Augmentation" vs. "Atrophy." We identified that uncritical adoption leads to **Cognitive Atrophy**, where students accumulate "cognitive debt." By bypassing the productive struggle of debugging and syntax generation, students fail to activate the Multiple Demand network required for deep conceptual understanding (Ivanova et al., 2020). This produces graduates who suffer from an illusion of competence, capable of prompting code they cannot verify, maintain, or secure.

7.3 Addressing RQ3: A Framework for Cognitive Apprenticeship 2.0

Finally, regarding the necessary pedagogical evolution (RQ3), we presented the Cognitive Apprenticeship 2.0 model. Through the pilot implementation in the USN Bachelor Thesis program and a technical course, we demonstrated that assessing the *process* of AI orchestration, rather than just the final code, provides a viable path to verify student competence in the post-GenAI era. To produce the "architects" and "orchestrators" the market now demands, educators must prioritize *Adaptive Expertise* (Hatano et al., 1986). This requires specific design principles:

1. **From Construction to Verification:** Assignments must shift focus from generating code to auditing, debugging, and integrating AI-generated outputs.
2. **Process-Oriented Assessment:** The adoption of Interactive Oral Assessments (IOA) and version history forensics to evaluate the *mental model* rather than the final artifact.
3. **Critical AI Literacy:** Teaching the material realities, ethical limitations, and "hallucination" risks of LLMs to prevent technological determinism.

7.4 Current Limitations and Future Research

While this paper argues for an urgent restructuring of software engineering education and presents an initial implementation of the Cognitive Apprenticeship 2.0 framework, it is necessary to outline the boundaries of our analysis and the constraints of the current validation.

7.4.1 Dependence on "Gray Literature" and Real-Time Data

A primary limitation of the labor market analysis is its reliance on non-peer-reviewed "gray literature," including industry reports and working papers. While this strategy is necessary to capture the real-time velocity of GenAI's impact, these sources lack the rigor of established economic census data. The "hollow middle" phenomenon should be interpreted as a strong *market signal* rather than a settled historical fact.

7.4.2 Scope of the Pilot Implementation

The implementation of the CA 2.0 framework described in this paper is currently limited to a single pilot cohort: the Spring 2026 Bachelor Thesis and the BID3000 course at the University of South-Eastern Norway. While the mandate for students to utilize and document GenAI usage mirrors corporate "AI Orchestration" workflows, the data collected is currently qualitative consisting of process logs, reflection notes, and oral defense performance. Furthermore, as this is a capstone project, the students have already completed the majority of their syntax-based training under traditional curricula. Therefore, this pilot validates the *Assessment* and *Orchestration* pillars of our framework but does not yet measure the impact of CA 2.0 on early-stage novices (freshmen) who have not yet formed mental models of code.

7.4.3 Need for Longitudinal Verification

The distinction between "Cognitive Augmentation" and "Cognitive Atrophy" may not be immediately visible in a single semester's output. High-quality thesis products produced with GenAI support may mask underlying gaps in foundational understanding that only manifest later in professional practice (the "brownfield" maintenance problem). Future research must track this specific cohort of graduates into their first two years of employment to verify if the "process reflection" mandated in their thesis successfully mitigated the risks of cognitive debt.

7.5 Final Remarks

The disappearance of the junior developer role is a crisis, but it is also an opportunity to elevate the discipline of computer science education. By abandoning the rote memorization of syntax in favor of deep systems thinking and ethical stewardship, we can prepare a generation of technologists who are

not merely "users" of AI, but its masters. The tools have changed, but the mission of the university remains constant: to cultivate the human capacity for critical thought in a world that increasingly seeks to automate it.

Generative AI Disclosure

The author used generative AI tools (ChatGPT 5.1, Claude Opus 4.5, and Gemini 3 Pro) during the preparation of this manuscript solely for language editing (clarity, grammar, and readability) and for high-level feedback on early draft structure and argumentation. These tools were not used to generate original research data, analyses, results, or conclusions. All substantive content, interpretation, and final editorial decisions were made by the author, who also reviewed and verified the manuscript for accuracy, completeness, and proper citation.

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